Unmasking the Masked: A Classical Machine Learning Pipeline for Detecting Forged Receipts

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***Abstract*—The abundance of digital and paper document forgery requires strong automated detection tools against financial fraud. The paper provides a classical machine learning method for forged receipt detection using multimodal features from image and text modalities. The approach entailed designing a feature set to obtain textural and statistical attributes from receipt images via Local Binary Patterns (LBP) and Canny edge detection, along with structural features obtained from the associated text files. Another demanding issue in this area is the excessive class imbalance between genuine and forged documents. To overcome this issue, Synthetic Minority Over-sampling Technique (SMOTE) was used to create a balanced training dataset. The models were assessed using the macro F1-score, precision, recall, PR AUC and ROC AUC to address class imbalance. The enhanced detection of the minority class was achieved through the use of SMOTE, while hyperparameter tuning led to improvements in the performance. The final Tuned Support Vector Machine model achieved a macro F1-score of 0.5429, and it had the highest recall on forged receipts, demonstrating it detected more histories of tampered documents effectively. This research sets a good baseline for receipt forgery detection and emphasizes that class imbalance solving is a key towards creating a working system.**

***Index Terms*—Document Forgery Detection, Local Binary Patterns, Machine Learning, Support Vector Machine (SVM)**

I. Introduction

The prevalence of digital imaging and advanced editing software has caused a tremendous boom in document falsification, which poses a very serious threat to companies and financial institutions. The most falsified documents by far are receipts, which are altered for expense report fraud, inflation of insurance claims, and tax fraud. The manual checking of such documents is time-consuming, costly, and prone to errors. This requires the creation of systems that can reliably and effectively discriminate between genuine and counterfeit receipts.

Creating such a system, though, is fraught with several difficulties. Forgeries are sometimes subtle and multifaceted and may range from straightforward text manipulations to sophisticated "Frankenstein" documents produced by combining pieces from various sources. In addition, one of the main challenges in this space is the natural **class imbalance** of real-world data: forged documents simply do not exist nearly as often as legitimate ones. This data bias can severely skew machine learning algorithms toward patterns based on the vast majority of the data and away from the limited number of rare instances they are meant to detect.

This work presents and assesses a traditional machine learning workflow for detecting receipt forgery, addressing these challenges directly. Our solution is multimodal, designing a collection of discriminative features from both visual properties of the receipt image and structural properties of the textual content of the receipt. To mitigate the issue of class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is used to generate a balanced training set. We compare a number of machine learning classifiers and show that this hybrid approach provides a strong baseline for the task of forgery detection.

The paper is organized as follows: the **Methodology** section describes the steps of data preprocessing, feature engineering, and model training procedures. The **Results** section reports the performance of the models. The **Discussion** section explains the outcomes and limitations observed in the research. Finally, the **Conclusion** section summarizes the study and outlines how the limitations can be addressed in future work.

II. Literature Review

Many researchers have tackled the issue of forgery detection in various ways, including through pattern-based analysis, deep learning and the classification of document features.

Ahsan et al(2024) [1] recognizes a major barrier to effective use of digital forgery detection systems as laboratory-to-practice differences based on differences in document formats and image quality. Similarly, Roja et al (2023) [2] describes the basic elements of digital image processing systems and demonstrates how pattern recognition drives technologies such as OCR and fingerprint identification—abilities fully transferable to scanned document forgery detection.

Prachi (2023) et al [3] discusses the application of deep learning approaches, particularly convolutional autoencoders (CAE) and convolutional neural networks (CNN), to identify-watermark patterns and other repeating structures. Her results demonstrate that deep models are precise but sensitive to input quality. Mohammed (2022) et al [4] also address this issue by highlighting the limitations of traditional image based methods and spectroscopic methods for addressing forgery. The authors propose the use of Laser-Induced Breakdown Spectroscopy

(LIBS) for non-destructive verification, and emphasizes the use of physical documents long term for critical events.

Pulkit (2023) et al [5] offers a systematic feature categorization of document security features—by type, inspection level, and security level—and examines their implementation in Indian identity documents. His research emphasizes the significance of feature-based categorization in the fight against document fraud, which can inform pattern-based detection approaches.

Chenfan (2023) et al [6] reviews current approaches to natural image tampering and generalizes them to the document field, proposing the Document Tampering Detector (DTD), a tool that combines spatial pattern analysis for more sensitive detection of faint irregularities. Although effective, his critique points to a lack of solidity in previous methods when applied across diverse document types.

Jhankar et al (2024) [7] proposes a multi-modal system that combines visuals, text, and metadata of documents to improve the accuracy of forgery detection. It notes that the one-feature systems are more prone to malicious attacks and format problems and suggests that deep architectures (such as ResNet-50) are better in complex modern types of documents.

Joel (2023) et al [8] adds seminal contributions to pattern recognition and deep learning. Joel classifies the learning approaches—supervised, unsupervised, semi-supervised, and reinforcement learning—and discusses the nature and applications of different architectures of deep neural networks. These concepts are indispensable for comprehending how deep learning models can be utilized successfully in scanned document analysis and forgery detection.

Pushpalata (2023) et al [9] and Xin (2023) et al [10] also add to this literature by concentrating on domain-specific datasets and enhancing training resilience, though those results are not investigated here in detail.

Kilari et al. (2024) [11] used CNN in their research to detect copy-move forgery in digitally modified identity documents. By applying the CNN on the MICC-F2000 datasets, they were able to get a training accuracy of 98.33%, 100% precision, and an F-score of 98%. When applied to a self-created dataset, IDCCMFD-1000, the CNN reported a training accuracy of 92% and a testing accuracy of 93.02%. These results demonstrate that the CNN model was easily transferable across different datasets.

In another study, Aboudramane et al. (2024) [12] present Doc-patch, an unsupervised method to detect forgeries in documents without the use of labeled training or testing data. Doc-patch uses FAISS K-Nearest Neighbors (KNN) to compare embeddings to a memory bank of verified authentic documents, and documents are identified as forged if their anomaly scores exceed a defined threshold. Doc-patch outperforms GANomaly, a popular unsupervised baseline, by a significant margin.

In their work, Naglaa et al. (2023) [13] presented the model IMD-CNN, using unique spectral signatures of the inks to facilitate identification and detection of document forgeries. This approach not only detects forged documents, but also determines the number of inks within the document, and detects forgeries by locating the regions that have been manipulated within a document. The results show that the model correctly identified forgeries with 99.91% accuracy for blue ink and 97.51% accuracy for black ink, surpassing previous work by 0.18% (blue ink) and 0.36% (black ink).

Teerath et al. (2022) [14] implemented another approach to overcome issues with the limited number of datasets available to researchers and the problems associated with annotation by developing a four-step algorithm for creating synthetic forged documents. The authors employed a fill-in-the-blank approach, where data is filled into documents with various characteristics, characters are randomly altered on different occasions by misalignment of the document or the addition of background noise, augmentation was applied to augment the collection of the synthetic forged document, and the authors would retain the bounding box information for the annotations into a JSON file.

A dataset was created by Beatriz et al. (2023) [15]—the same one utilized in this study—by conducting workshops where 19 compatriots of varying ability in editing created and annotated forgeries, and experimentation of the dataset reported that traditional methods such as Bag-of-Words with Logistic Regression (F1: 17.78%) and SVM on raw pixels (F1: 13.33%) performed poorly. In comparison, ChatGPT in a zero-shot condition was able to achieve greater recall but very low precision (F1: 28.39%). The best performance came from a CNN that was trained to detect JPEG compression artifacts, which attained around 78–79% precision.

III. Methodology

With the purpose of developing a strong classification model to differentiate forged receipts from authentic ones, traditional machine learning techniques were employed, with feature engineering from both image and text sources. The main steps consist of data preprocessing, multimodal feature extraction, handling major class imbalance, model training, and thorough evaluation of the models.

A. Dataset Structure

The dataset used in this study is the Find it again! dataset [16], obtained from Kaggle. The dataset contains receipt images in `.png` format and corresponding extracted text in `.txt` format, obtained using Optical Character Recognition (OCR). In addition, the dataset was divided into training, validation, and test sets, which are specified in the files `train.txt`, `val.txt`, and `test.txt`, respectively. This division ensures that the models do not carry forward any bias from the training data.

B. Feature Engineering

For the feature set, both visual and structural properties of the receipts were considered and extracted.

1) Image Feature Extraction: Visual evidence often serves as a primary forgery indicator. To extract such information, features were computed from the grayscale version of each receipt image resized to a standard 256×256 pixel size.

- **Texture Analysis:** Local Binary Patterns (LBP) was used to examine the textural characteristics of the receipts. LBP defines image pixels by thresholding neighboring pixels and encoding results as binary values. A 10-bin histogram of the LBP result was calculated to produce a compact texture feature vector.
- **Statistical Features:** Basic image statistics, including the mean and standard deviation of pixel intensities, were calculated to capture information about contrast and brightness.
- **Edge Density:** The Canny edge detector was applied to each image to identify edges. Edge density, defined as the number of edge pixels divided by the total number of pixels, was used as a feature to detect irregular patterns or artificial lines introduced through tampering.

2) Text Feature Extraction: The layout and wording of the text on a receipt may also provide clues of tampering. The following text-based features were extracted:

- **Token Count:** Total number of words.
- **Digit Count:** Total number of numeric digits.
- **Average Token Length:** Mean length of tokens.
- **Keyword Presence:** Binary indicators of whether essential keywords such as "total" and "date" are present, which are fundamental to a valid receipt.

These image and text features were concatenated to form a single feature vector per sample.

C. Managing Class Imbalance via SMOTE

Initial exploration revealed a significant class imbalance between genuine and forged receipts, with the authentic class dominating. Training directly on such imbalanced data risks producing models biased toward the majority class. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied.

SMOTE creates synthetic samples of the minority class by randomly selecting an instance, finding k nearest neighbors, and generating a synthetic sample along the line segment joining the instance and one neighbor. This augmentation was applied only to the training set, ensuring no information leakage into validation or test sets.

D. Training Models and Tuning Hyperparameters

Two approaches were taken for training:

1) Training on original data with class weight adjustments.
2) Training on the SMOTE-balanced dataset.

The four algorithms evaluated were:

1) Logistic Regression (a basic linear classifier).
2) Random Forest (an ensemble of decision trees).
3) Support Vector Machine (SVM), which finds a separating hyperplane.
4) XGBoost, a gradient-boosting framework.

For raw-data models, class weighting parameters such as `class_weight='balanced'` (Logistic Regression, Random Forest, SVM) and `scale_pos_weight` (XGBoost) were used to mitigate class imbalance. For SMOTE-balanced models, a `StandardScaler` was applied to normalize training data, which was later used to transform validation and test sets.

After preliminary evaluation, the models showing the best performance (Random Forest, XGBoost, and SVM) from the SMOTE pipeline were re-evaluated for hyperparameter optimization. Randomized Search Cross-Validation (`RandomizedSearchCV`) was employed, optimizing for the macro F1-score to achieve balanced performance across both classes.

E. Evaluation Metrics

Model performance was assessed using a comprehensive set of metrics suitable for imbalanced classification:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where
TP- True Positive
TN- True Negative
FP- False Positive
FN- False Negative

- **Precision (Positive Predictive Value):**

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

- **Recall (Sensitivity):**

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

- **F1-Score:** Harmonic mean of Precision and Recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

The *macro F1-score* is the unweighted mean of F1-scores across classes:

$$F1_{macro} = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{5}$$

where C- Number of classes.

- **ROC AUC:** Area under Receiver Operating Characteristic curve, where ROC plots True Positive Rate (TPR) vs. False Positive Rate (FPR):

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{6}$$

- **PR AUC:** Area under the Precision–Recall curve, which plots Precision vs. Recall and is particularly informative for imbalanced datasets as it emphasizes performance on the minority (forged) class.
- **Precision, Recall, and F1-Score:** Computed for both "Authentic" and "Forged" classes. The macro F1-score

was chosen as the primary comparison metric, as it balances precision and recall while mitigating the effect of imbalance.

Finally, models were compared in a consolidated table, and the highest-performing model was selected based on the macro F1-score on the held-out test set. The overall pipeline of the proposed methodology is illustrated in Fig. 1.

IV. Results

In this section, we present the empirical results of our experiments. We compare the performance of four machine learning models under three different conditions: (1) training on original imbalanced data, (2) training on SMOTE-balanced data, and (3) training the best-performing models on SMOTE-balanced data with hyperparameter optimization. The macro F1-score is used as the main metric for model comparison, as the dataset is highly imbalanced.

A. Baseline Performance on Imbalanced Data

The first evaluation considered the performance of models trained on the original imbalanced dataset. Class weighting techniques were employed to handle the imbalance.

The results in Table I show that despite relatively high accuracy, the models were unable to detect the minority class ("Forged"). For example, the Random Forest model achieved an accuracy of 83.49% but completely failed to identify forged receipts (recall = 0.00). Logistic Regression and XGBoost also performed poorly in terms of recall and F1-score for the forged class, highlighting the limitations of class weighting alone.

B. Impact of SMOTE on Model Performance

To improve detection of the minority class, models were retrained on data balanced using SMOTE.

As shown in Table I, nearly all models trained with SMOTE exhibited improved recall and F1-scores for the forged class compared to those trained without SMOTE. For example, the Random Forest model's F1-score improved from 0.4550 to 0.5111, and it was able to correctly classify some forged documents (recall = 0.1429). Similarly, the SVM model improved from an F1-score of 0.4003 to 0.5035. These results demonstrate the effectiveness of SMOTE in enhancing classifier balance.

C. Selection of the Final Model and Hyperparameter Tuning

The three best-performing models from the SMOTE experiments—Random Forest, XGBoost, and SVM—were further tuned using Randomized Search Cross-Validation.

Among these, the tuned SVM showed the most notable improvements, with its F1-score increasing from 0.5035 to 0.5429 and PR AUC rising from 0.1835 to 0.2464. Based on these results, the tuned SVM trained on SMOTE-balanced data was selected as the best-performing model. It achieved the highest macro F1-score (0.5429), the highest PR AUC (0.2464), and the highest recall for the "Forged" class (0.3143), indicating the most balanced and effective performance across all models.

D. Visual Analysis of Performance

The quantitative results are further supported by the Precision-Recall (PR) curves in Figure 2. Models trained with SMOTE (right plot) show superior curves compared to those trained without SMOTE (left plot). The curve for the tuned SVM (green line, AUC = 0.2464) is the most prominent, visually confirming its superiority for this classification task.

V. Discussion

The findings suggest that a classical machine learning framework, enhanced by multimodal feature engineering and effective class balancing techniques, can be useful for receipt forgery detection. This section interprets the key findings, highlights limitations, and proposes directions for future work.

A. Summary of Key Findings

The experiments yielded several important insights, particularly regarding the necessity of addressing class imbalance.

The baseline models performed poorly despite the use of class weighting. For instance, the Random Forest classifier achieved an overall accuracy of 83.49%, but its recall for the "Forged" class was zero. Thus, it provided no value for the primary task of detecting forgeries. This result exemplifies one of the pitfalls of classification on imbalanced datasets: accuracy can be misleading when one class dominates the distribution.

On the positive side, applying SMOTE improved the performance of all models, particularly in terms of F1-score and PR AUC, which are more appropriate measures for imbalanced problems. By generating synthetic minority samples, SMOTE enabled the models to better capture the distinctive characteristics of forged receipts instead of defaulting to the "Authentic" class. This demonstrates that balanced training data is essential for forgery detection, and that class weighting alone is insufficient.

The multimodal feature set proved effective, highlighting the value of incorporating both visual and textual features. Visual features such as LBP and edge density capture artifacts of tampering and irregular textures, while textual features such as token counts and keyword presence reveal logical inconsistencies introduced during electronic edits. The combination of these features created a richer representation of receipts, making forgery detection more robust.

Finally, the tuned SVM emerged as the best-performing model. This suggests that SVMs are particularly well-suited for this task due to their ability to find optimal non-linear decision boundaries in the engineered feature space. Although ensemble models like Random Forest and XGBoost are powerful, they did not capture the subtle differences between genuine and forged receipts as effectively as the tuned SVM.

B. Limitations of the Study

While the approach was successful, several limitations remain:

- **Reliance on Handcrafted Features:** The methodology depends on manually engineered features. This approach

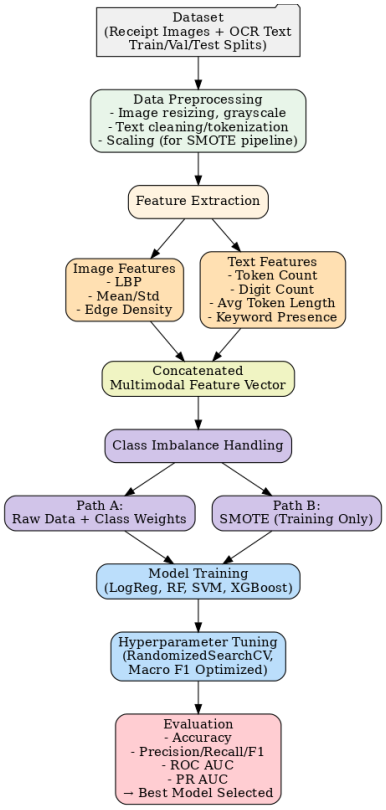


Fig. 1. Proposed methodology workflow for forged receipt detection, including preprocessing, feature extraction, class imbalance handling, model training, and evaluation.

TABLE I
COMPREHENSIVE MODEL COMPARISON ON THE TEST SET

Model	SMOTE	Accuracy	F1-Score	PR AUC	ROC AUC	Precision (Forged)	Recall (Forged)
Logistic Regression	Without	0.4862	0.4108	0.1436	0.4842	0.1333	0.4000
Random Forest	Without	0.8349	0.4550	0.1663	0.5006	0.0000	0.0000
SVM	Without	0.4358	0.4003	0.1393	0.4554	0.1615	0.6000
XGBoost	Without	0.7936	0.5319	0.1635	0.4507	0.2500	0.1429
Logistic Regression	With	0.4817	0.4201	0.1401	0.4681	0.1518	0.4857
Random Forest	With	0.7615	0.5111	0.1845	0.5727	0.1852	0.1429
SVM	With	0.6101	0.5035	0.1835	0.5909	0.1951	0.4571
XGBoost	With	0.7431	0.5121	0.1884	0.5803	0.1818	0.1714
Random Forest (Tuned)	With	0.7661	0.5269	0.1820	0.5702	0.2143	0.1714
XGBoost (Tuned)	With	0.7339	0.5176	0.1995	0.6023	0.1892	0.2000
SVM (Tuned)	With	0.7156	0.5429	0.2464	0.6061	0.2245	0.3143

may be less powerful than end-to-end deep learning models (e.g., CNNs), which can learn complex representations directly from raw data.

- **Dependence on OCR Quality:** The accuracy of text-based features depends on the quality of the OCR process. Errors or noise in OCR output can significantly reduce detection performance in real-world settings.
- **Limited Feature Scope:** The feature set, while effective, remains relatively simple. Advanced techniques such as semantic NLP analysis, EXIF metadata inspection, or Error Level Analysis (ELA) were not explored.
- **Dataset Generalizability:** The model was trained and tested on a single dataset. Its performance may not generalize to receipts from other domains, languages, currencies, or forgery techniques. Real-world forgeries may differ significantly from those in the dataset.

VI. CONCLUSION AND FUTURE WORK

This study highlights the possibility of using a standard machine learning pipeline for detecting forged receipts using multimodal features from both the images and text. An important finding was that addressing class imbalance is crucial, where class weighting was used. On the other hand, using SMOTE helped improve minority class representation. The Support Vector Machine trained with hyperparameters achieved the best performance of all models with a macro F1-score of **0.5429**, which is an informative benchmark for future work.

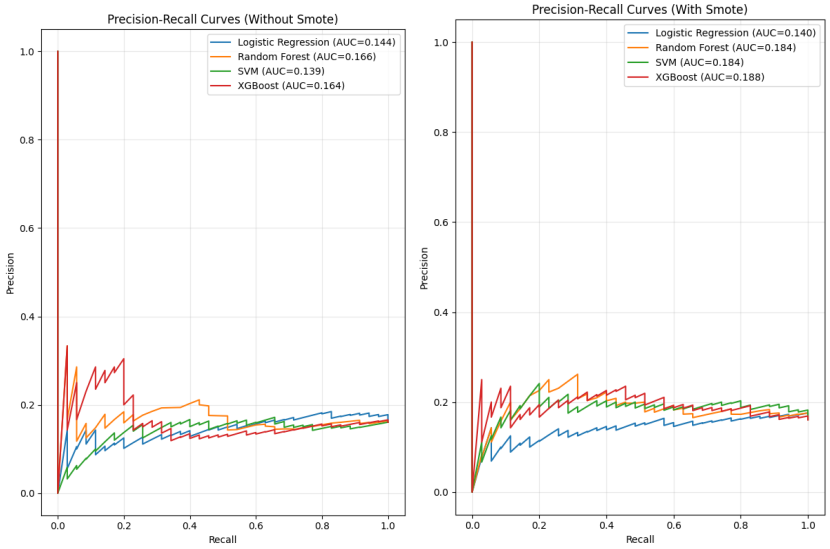


Fig. 2. Precision-Recall curves comparing models trained without SMOTE (left) and with SMOTE (right). The Area Under the Curve (AUC) values are shown in the legend.

For future work, there are several directions to further expand this line of work: (i) build end-to-end deep learning architectures that integrate convolutional neural networks (CNNs) for extracting features from images along with transformer-based models for text, (ii) use OCR integrated with forgery detection to allow for noise present at the character level, (iii) develop a dataset composed of various formats and types of forgery, and (iv) study forensic features like Error Level Analysis and font consistency. These avenues could generate higher quality representations and more robust generalizable models for detection of document forgery.

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